Regularisation and early stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

If validation chooses an earlier checkpoint, which parameters should be used after selection?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Separate a weight penalty from prediction error. Try the recall before opening notes.

Your learning target

I can separate a **weight penalty** from **prediction** error.

I can trace a regularised update under a stated objective.

I can apply a stopping rule and **restore** the best observed checkpoint.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Separate a **weight penalty** from **prediction** error

Trace a regularised update under a stated objective

Apply a stopping rule and **restore** the best observed checkpoint

Retrieve what you already know

> If validation chooses an earlier checkpoint, which parameters should be used after selection?

My starting explanation

This week's end goal

By the end, I can explain **regularisation** and early stopping using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Regularisation and early stopping case. Separate observations, given facts and assumptions before choosing an explanation.

Keep h = ReLU(x), y = vh, train only v on (1, 2) and (2, 4), and freeze all hidden parameters and the zero **output** bias. **Training** MSE is L(v) = 2.5(v - 2) squared.

Define J(v) = L(v) + lambda times v squared, with lambda nonnegative. Only this one **output** weight is penalised here.

For lambda = 0.5, the penalty at v = 1.5 is 0.5 times 2.25 = 1.125. **Prediction** MSE is 0.625, so total objective J is 1.75. Keep these quantities separate in logs.

A smaller total objective does not automatically mean a smaller **prediction** error.

v	Prediction MSE	Penalty at lambda 0.5	Total J
1	2.5	0.5	3
1.5	0.625	1.125	1.75
2	0	2	2

Among these three candidates, the smallest penalised objective is at v = 1.5, whereas the smallest **training** MSE is at v = 2. Penalising a weight changes the tradeoff.

It does not guarantee better validation performance or fair outcomes.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Regularisation and early stopping. Connect each labelled input, deciding step and result.

The data gradient is 5(v - 2). The penalty gradient is 2 times lambda times v, giving total gradient 5(v - 2) + 2 lambda v. At v = 1.5 and lambda = 0.5, these contributions are -2.5 and 1.5, so total gradient is -1.

A rate 0.1 step gives v = 1.6.

```
def terms(v, penalty):
    data_loss = 2.5 * (v - 2) ** 2
    weight_cost = penalty * v * v
    gradient = 5 * (v - 2) + 2 * penalty * v
    return data_loss, weight_cost, data_loss + weight_cost, gradient

v = 1.5
penalty = 0.5
old = terms(v, penalty)
v = v - 0.1 * old[3]
print(round(v, 6))  # 1.6
print([round(x, 6) for x in terms(v, penalty)])
# [0.4, 1.28, 1.68, -0.4]
```

Assume finite numeric values and nonnegative penalty. At the new v, data loss is 0.4, penalty 1.28 and total 1.68.

Choose penalty strength using a declared validation **prediction** metric, not by comparing total objectives with different lambdas as if they were the same measure.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Regularisation and early stopping. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Change **patience** from 1 to2 under the same tie rule.

> **Predicted result**
>
> **Prediction**: Stop at checkpoint 4; still **restore** checkpoint 2.

> **Why the result follows**
>
> Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Regularisation and early stopping to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Regularisation	A constraint or training procedure intended to limit fitting choices and improve generalisation
Weight penalty	An added objective term that discourages specified large parameter values
Patience	The allowed run of consecutive non-improving evaluations before stopping under a stated rule
Restore	Reload the parameters saved at the selected checkpoint

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Regularisation and early stopping, predict a step, then compare with its explanation.

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Regularisation and early stopping

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

The facts we will use

Supplied facts

Toy objective $J(a)=(a-3)^2+2a^2$; start $a = 0$; gradient $6a-6$; rate 1/6. Separate validation losses: [5, 3, 3, 4], **patience** 1, no ties as improvement.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/16

Regularisation and early stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regularised update

At 0 gradient -6; update gives a = 1.



2. Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.



3. Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

CHANGE ONE THING

Change patience from 1 to2 under the same tie rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regularised update	At 0 gradient -6; update gives a = 1.
2. Split objective	At 1 prediction term 4, penalty 2, total 6; at 0 total 9.
3. Early-stop separately	Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

What this example does not establish

Keep the stated limit

Regularisation and early stopping are different tools; neither guarantees better

performance in every dataset.

Keep the stated limit

Read the labelled picture: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change **patience** from 1 to2 under the same tie rule. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Regularisation and early stopping claim and explain a repair using the supplied evidence.

Check the claim against the facts

Early stopping limits the **training** duration using development **evidence**; it does not add a weight term to the objective. For this policy, initialise the best checkpoint to epoch 0 and the miss count to zero.

After each later evaluation, save and reset only on a strict improvement; otherwise increment the count. Stop as soon as the count reaches **patience**, then reload the best checkpoint.

Check the claim against the facts

Suppose a pre-recorded demonstration also lists epoch 7 loss 1.3. A live run following the rule has already stopped at 6 and does not observe 7. Do not use future values to rewrite the stopping decision.

The policy can miss a later improvement; **patience** balances waiting against a **training** budget and must be specified in advance.

Check the claim against the facts

Save actual model parameters with the checkpoint, not just its epoch number or loss. With a fixed maximum epoch budget, stop when either limit is reached and **restore** the best observed state.

An exact tie keeps the earlier checkpoint under our rule. Different minimum-improvement thresholds are possible, but would be different declared policies.

Locate the earliest unsupported step

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

State one repair and the checks you would rerun.

Practise together

Practise Regularisation and early stopping with support, then explain the deciding step in your own words.

At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the next update

Use workbook W2 for the connected example "**Regularisation** and early stopping". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Regularisation** and early stopping. Proposed explanation: “Zero **prediction** error means no regularised update”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Regularisation and early stopping. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

```
def terms(v, penalty):
    data_loss = 2.5 * (v - 2) ** 2
    weight_cost = penalty * v * v
    gradient = 5 * (v - 2) + 2 * penalty * v
    return data_loss, weight_cost, data_loss + weight_cost, gradient

v = 1.5
penalty = 0.5
old = terms(v, penalty)
v = v - 0.1 * old[3]
print(round(v, 6))  # 1.6
print([round(x, 6) for x in terms(v, penalty)])
# [0.4, 1.28, 1.68, -0.4]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Regularisation and early stopping. Explain why each step is justified.

Explain → apply → challenge

> At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **regularisation** and early stopping **evidence** sheet to Image-pattern Network Investigation. Show how you can separate a **weight penalty** from **prediction** error.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Weight penalties and early stopping can be used together, but they act differently.

Record which weights receive penalties, the objective convention, lambda, learning rate, checkpoint frequency, **patience**, tie rule and maximum budget.

For this classroom model there are no extra learned pre**process**ing or running statistics; larger models may require more state to reproduce evaluation.

> **A team stops at epoch 6, uses its final parameters instead of the saved best state, and chooses patience by looking at final test losses. Identify the two mistakes and explain how to repair the experiment.**

Select these settings using validation, freeze the complete procedure, and reserve untouched final test data for reporting. Neither **regularisation** nor stopping makes a weak validation set representative.

Next week studies initialisation and reproducibility.

A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Regularisation and early stopping case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer both rules independently

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Regularisation and early stopping, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

By the end, I can	How I will show it
Separate a weight penalty from prediction error	Explain the deciding step, show a trace or test, and state one limit.
Trace a regularised update under a stated objective	Explain the deciding step, show a trace or test, and state one limit.
Apply a stopping rule and restore the best observed checkpoint	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Regularisation and early stopping without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate a **weight penalty** from **prediction** error

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

1. Split objective

2. Early-stop separately

3. Regularised update

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Early-stop separately Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change **patience** from 1 to2 under the same tie rule. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Stop at checkpoint 4; still **restore** checkpoint 2.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop at checkpoint 4; still **restore** checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Repair a missing building block

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

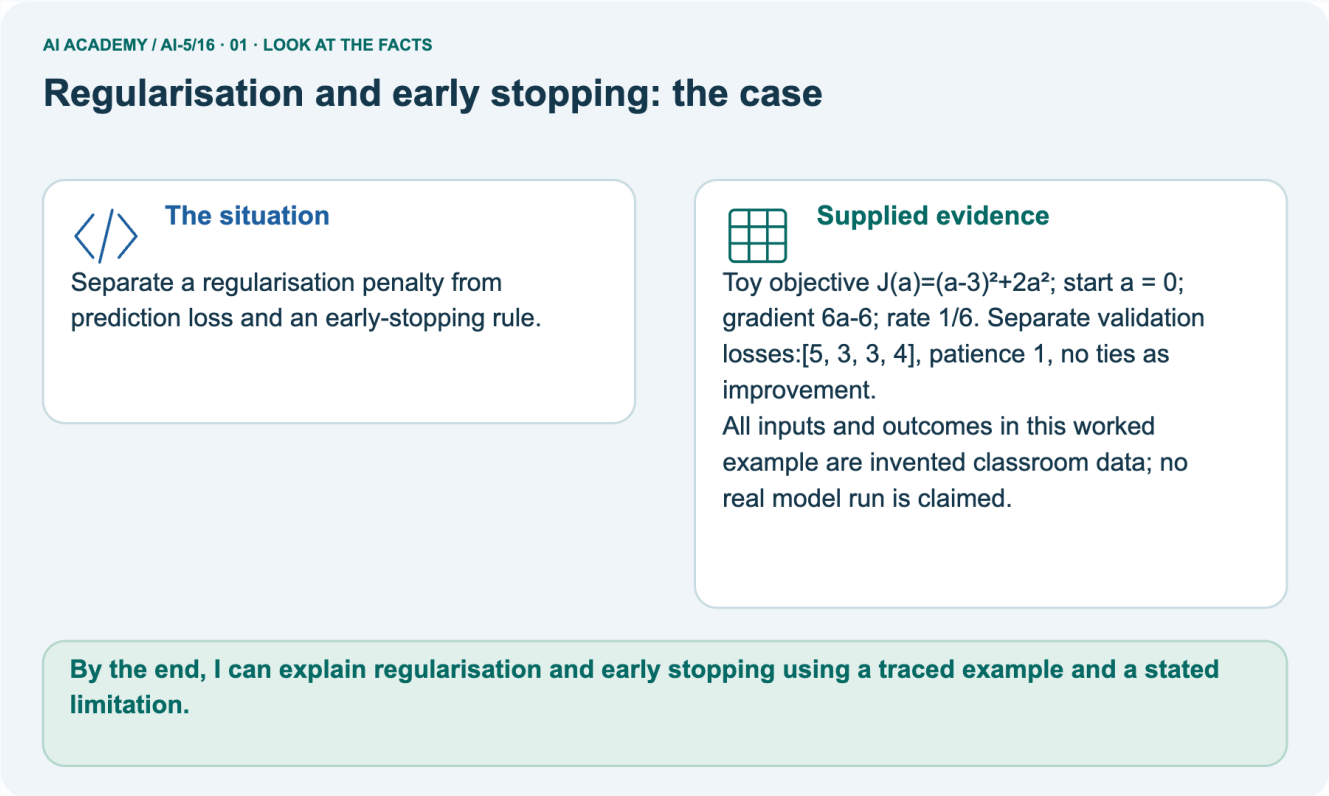
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Regularisation and early stopping graphic using its labels, deciding evidence and relevant limitation.

01. Regularisation and early stopping: the case



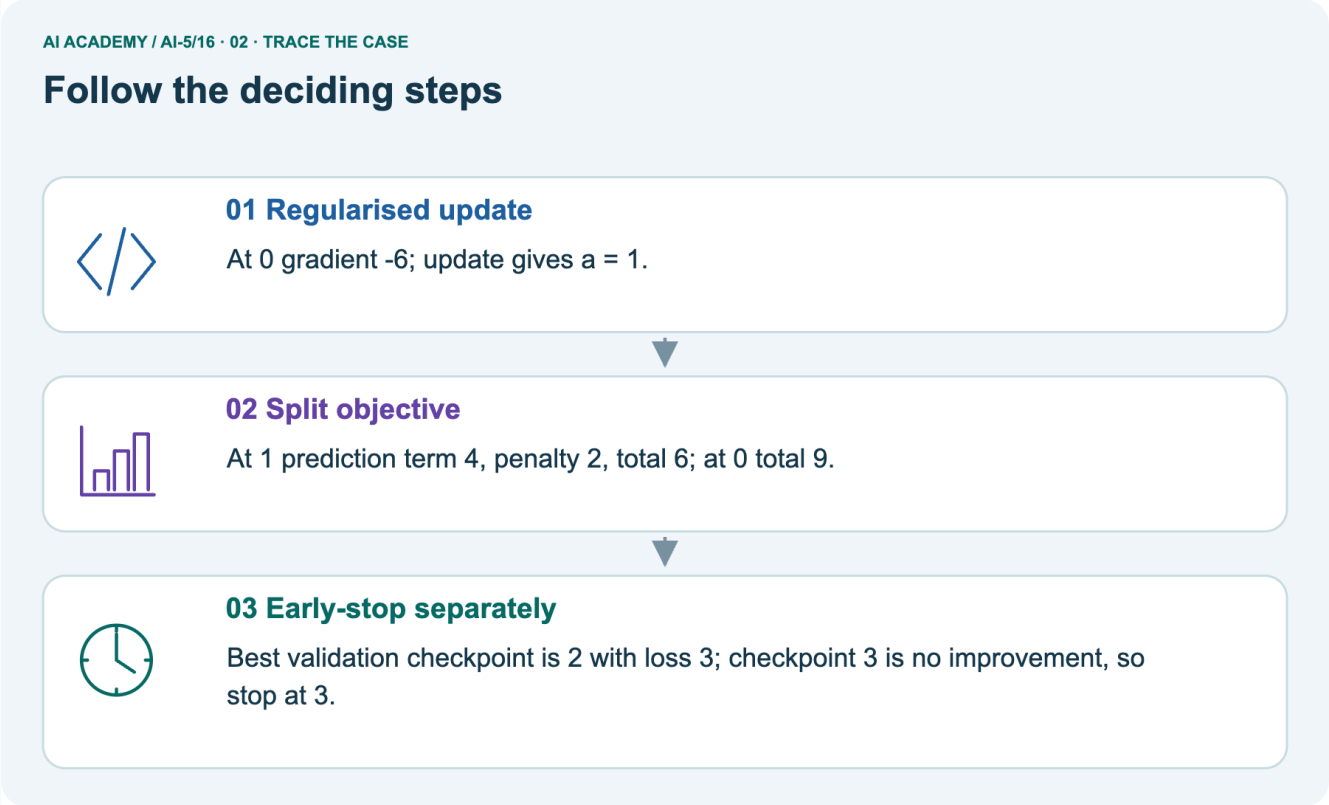
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule. Supplied facts: Toy objective J(a)=(a-3)²+2a²; start a = 0; gradient 6a-6; rate 1/6.

Separate validation losses:[5, 3, 3, 4], **patience** 1, no ties as improvement. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

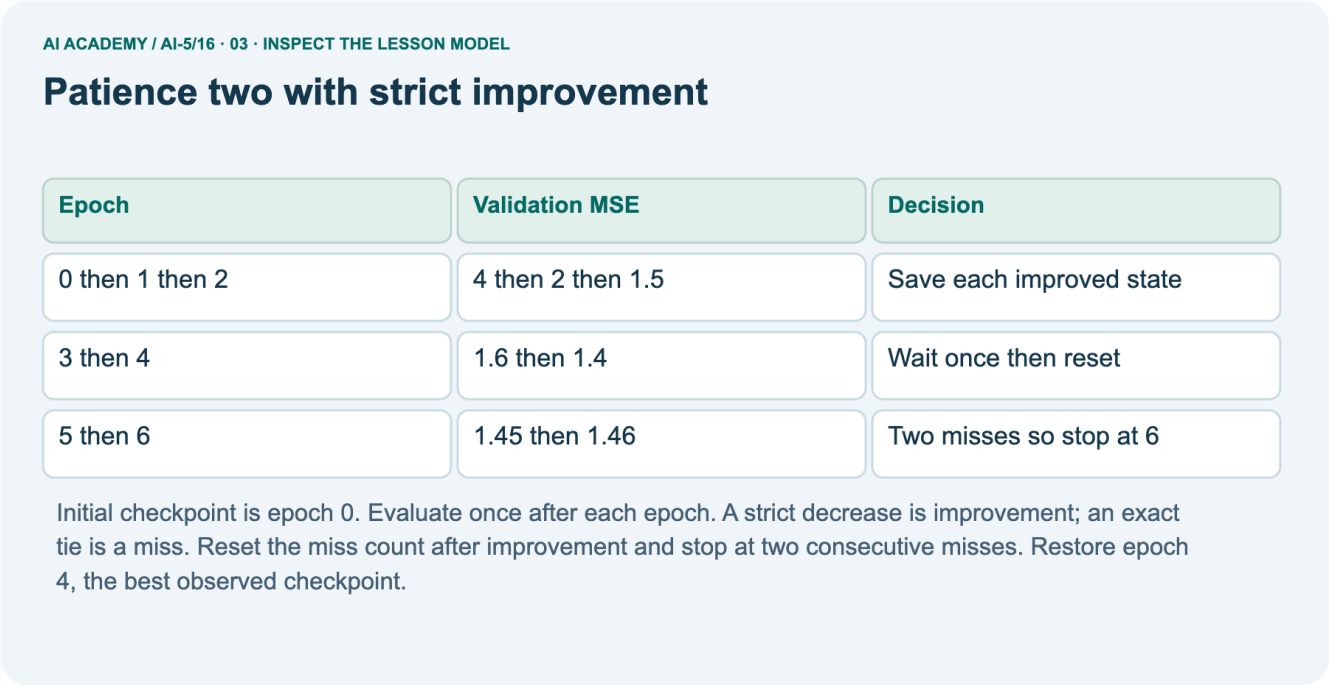


Follow the labelled picture with the matching explanation. Open the vector diagram

Regularised update: At 0 gradient -6; update gives a = 1. → Split objective: At 1 **prediction** term 4, penalty 2, total 6; at 0 total 9. → Early-stop separately: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Patience two with strict improvement

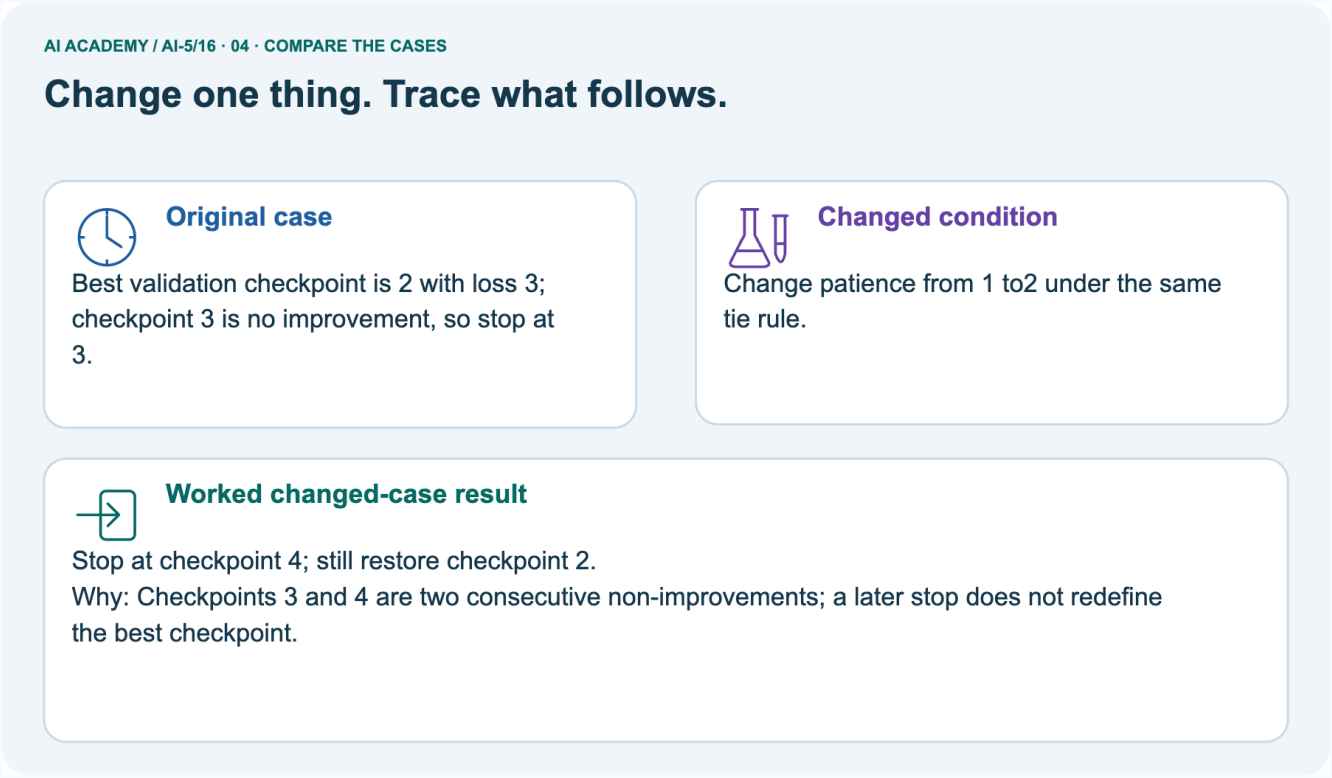


Follow the labelled picture with the matching explanation. Open the vector diagram

Patience two with strict improvement. 0 then 1 then 2 | 4 then 2 then 1.5 | Save each improved state. 3 then 4 | 1.6 then 1.4 | Wait once then reset. 5 then 6 | 1.45 then 1.46 | Two misses so stop at 6

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



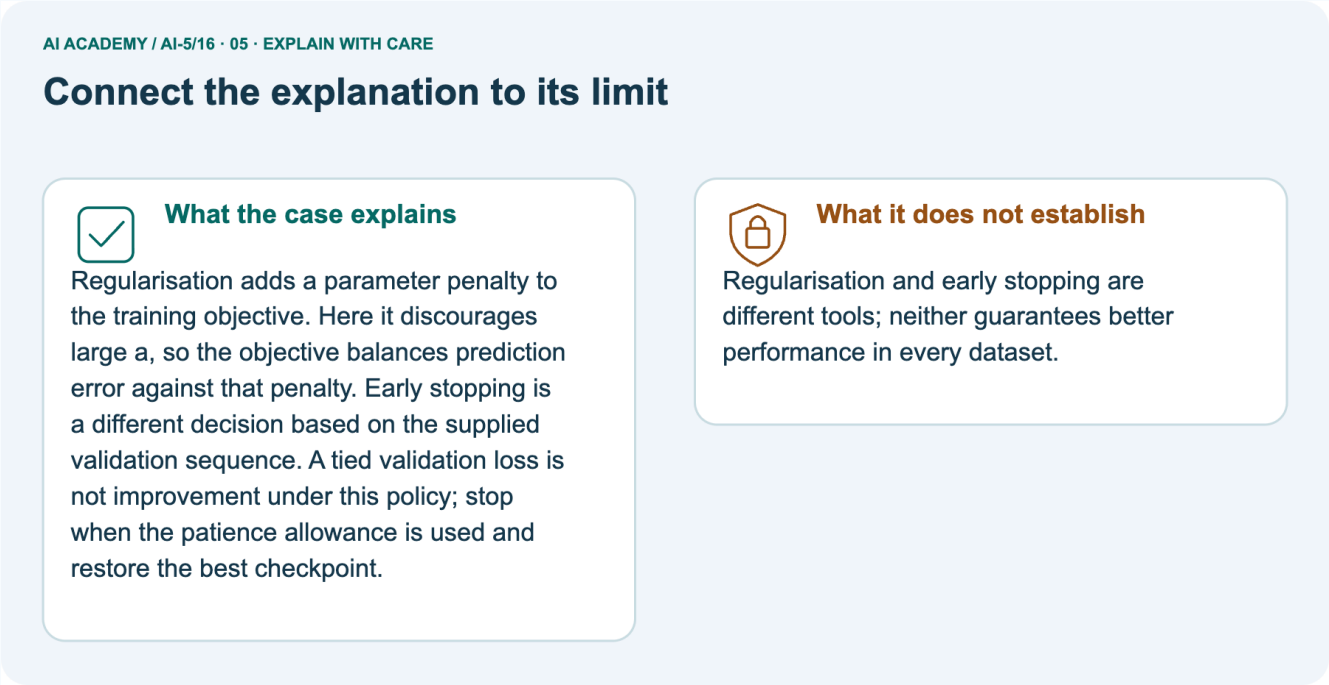
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3. Changed condition: Change **patience** from 1 to2 under the same tie rule.

Expected result: Stop at checkpoint 4; still **restore** checkpoint 2. Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

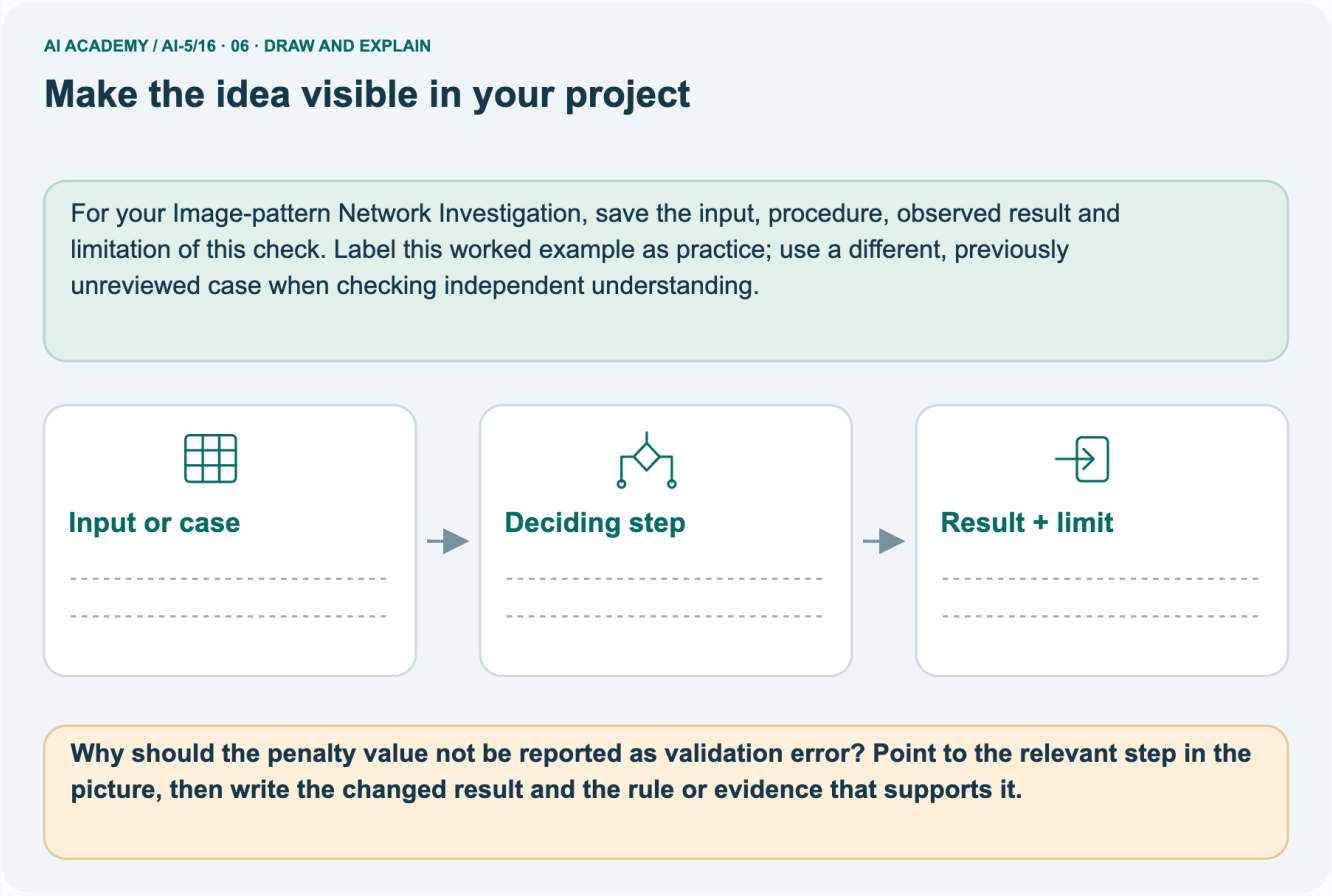
Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Limit: **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.